

Yajya Arora

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Summary

- **Third-year CS student** with a strong foundation in **full-stack web development**, **real-time systems**, and applied machine learning.
- Experienced building full-stack, real-time applications with **React**, **Node.js**, **Express**, **TypeScript**, **Socket.io**, **PostgreSQL**, and **Prisma ORM**, including layered backend architecture and **LLM API integration**.
- Proficient in **Python**, **NumPy**, **Pandas**, **Matplotlib**, and **Django REST Framework** for data analysis and statistical modeling.
- Solid grasp of **machine learning fundamentals**: linear/logistic regression, feature engineering, probability distributions, and statistical inference.
- Strong in **Data Structures & Algorithms** with **180+ problems solved on LeetCode**.
- Additional experience in **C**, **C++**, **R**, and **MATLAB**.

Education

Thapar Institute of Engineering and Technology

B.E., Computer Science and Engineering

CGPA: 8.7/10

Patiala, India

Sep 2024 – Present

Relevant Coursework

AWS Academy Cloud Foundations

Amazon Web Services (AWS)

Present

Projects

Loopin—AI-Powered Real-Time Collaborative Workspace | *Group Project*

- Built a **full-stack real-time collaboration platform** with task management, live notes, team messaging, and whiteboarding, backed by a layered backend architecture shared across REST and WebSocket entry points.
- Designed a **Socket.io messaging system** with an **AI agent layer** that generates daily stand-up summaries, decomposes tasks, and diagnoses blockers from project activity.
- Built a **native C++/Qt6/QML desktop client** using GitHub OAuth and libgit2 to monitor repos and generate AI-assisted commit messages.
- Collaborated in a **four-person team**, resolving cross-branch schema and dependency conflicts across a feature-branch workflow.
- *Tech Stack: React, Node.js, Express, TypeScript, PostgreSQL (Supabase), Prisma, Socket.io, C++, Qt6, QML*

AI-Driven Proactive Thermal Management for Cloud Infrastructure | *Developer, GitHub* Feb 2026 – Mar 2026

- Built a **predictive ML cooling system** to anticipate heat spikes and address **15–30% performance loss** associated with thermal throttling.
- Engineered **physics-aware time-series features** using Newton's Cooling Law, lag values, and rolling statistics on **15,757 telemetry samples**.
- Implemented a **Random Forest** model achieving $R^2 = 0.9516$ and **MAE $\approx 2.37^\circ\text{C}$** , outperforming the other evaluated algorithms.
- Enabled proactive **fan tuning and VM migration** before overheating to improve efficiency, stability, and hardware lifespan.
- *Tech Stack: Python, scikit-learn, pandas, Random Forest, Arduino Uno, Physics-Aware ML, Time-Series Analysis*

Skills

Databases: Supabase, PostgreSQL

Frameworks & Libraries: React, Node.js, Express, Prisma, Socket.io, Django REST, NumPy, Pandas, Scikit-learn

APIs & Real-Time Systems: REST APIs, WebSockets, Socket.io, GitHub OAuth, LLM APIs

Machine Learning: Supervised Learning, Probability Distributions, Statistical Inference, Feature Engineering

Programming Languages: C, C++, Python, JavaScript, TypeScript, SQL, R, MATLAB

Tools & Platforms: Git, GitHub, Arduino, VS Code, Jupyter Notebook, LeetCode, Codeforces, Docker, Supabase

Core: DBMS, Computer Networks, Object-Oriented Programming (OOP), DSA